

Peng, Jian-E., and Yuanlan Jiang. "Mining Opinions on LMOOCs: Sentiment and Content Analyses of Chinese Students' Comments in Discussion Forums." *System (Linköping)*, vol. 109, 2022, pp. 102879-, <https://doi.org/10.1016/j.system.2022.102879>.

This study explores opinion mining techniques applied to Language Massive Open Online Courses (LMOOCs), focusing on how sentiment analysis and natural language processing (NLP) can be leveraged to extract student feedback and enhance instructional design. The research highlights the effectiveness of models like TF-IDF and LDA for topic modeling, stating that "latent themes within discussion forums can provide valuable insights into student engagement and course effectiveness." The study also emphasizes the importance of instructor intervention, noting that "early identification of negative sentiment trends can help instructors address learner concerns proactively." This aligns with my research goals of improving discussion forum engagement through NLP-driven analysis, supporting my team's objective of designing a React-based dashboard to visualize student sentiment and discussion trends.

Jelodar, H., Wang, Y., Orji, R., & Huang, S. (2020). *Deep Sentiment Classification and Topic Discovery on Novel Coronavirus or COVID-19 Online Discussions: NLP Using LSTM Recurrent Neural Network Approach. IEEE Journal of Biomedical and Health Informatics*, 24(10), 2733-2742.

This study examines sentiment classification and topic modeling in online COVID-19 discussions, leveraging Latent Dirichlet Allocation (LDA) for topic discovery and Long Short-Term Memory (LSTM) networks for deep sentiment analysis. The researchers processed 563,079 comments from Reddit and found that LSTM outperformed traditional machine-learning classifiers (e.g., SVM, Naïve Bayes, Logistic Regression) with an accuracy of 81.15%. They highlight that "LDA is effective in detecting latent topics in large-scale discussion data," while LSTM can classify sentiment with higher precision than conventional models. The study also underscores the importance of removing noise and stop-words for improved semantic extraction and text classification. These findings align with the objectives of my research, as LDA and sentiment classification models could be applied to discussion forum datasets to extract insights on student engagement and instructional trends.

- Deep learning LSTM for sentiment classification can be an alternative to bert

Liu, Sannyuya, et al. "Automated Detection of Emotional and Cognitive Engagement in MOOC Discussions to Predict Learning Achievement." *Computers and Education*, vol. 181, 2022, pp. 104461-, <https://doi.org/10.1016/j.compedu.2022.104461>.

This study presents an automated model using BERT-CNN to detect emotional and cognitive engagement in online discussions, specifically within MOOC forums. The research finds that BERT-CNN outperforms traditional machine learning models, improving engagement recognition accuracy by 10% for emotional engagement and 8% for cognitive engagement. The study also highlights the importance of emotional engagement, stating that "positive and confused emotions contribute more to higher-level cognition than negative emotions" and that "co-occurring emotion and cognition indicators jointly predict learning

achievement with higher reliability". The authors emphasize the need for interpretable NLP models, addressing a common criticism of deep learning models in education. The research suggests that combining BERT with CNN enhances feature fusion, making the model more effective for classifying discussion posts in online learning environments.

- Bert is effective for classifying discussion posts
- Frameworks engagement analysis, can be applied to student discussion data in project
- The BERT-CNN approach could be tested against other models like TF-IDF or LDA to determine replicability.

Cummins, J. A., & Nambudiri, V. E. (2022). *Natural language processing: A window to understanding skincare trends. International Journal of Medical Informatics, 160*, 104705.
<https://doi.org/10.1016/j.ijmedinf.2022.104705>

This study explores the application of Latent Dirichlet Allocation (LDA) and sentiment analysis to analyze public discourse on skincare topics using Reddit discussions from the r/SkincareAddiction subreddit. The researchers processed over 3 million comments and used LDA to categorize 25 discussion topics while applying VADER sentiment analysis to gauge public sentiment over time. The study highlights LDA's effectiveness in sorting unstructured social media text into meaningful themes, stating that "LDA trained on skincare-related social media can successfully sort comments into medically relevant categories." The study also emphasizes that comment frequency analysis can reveal trends in public discourse, demonstrating a shift in terminology preferences over time. The authors acknowledge limitations in LDA's unsupervised approach, noting that "manual topic labeling is necessary as the model is not able to self-label topics."

Zou, W., Hu, X., Pan, Z., Li, C., Cai, Y., & Liu, M. (2021). Exploring the relationship between social presence and learners' prestige in MOOC discussion forums using automated content analysis and social network analysis. *Computers in Human Behavior, 115*, 106582. <https://doi.org/10.1016/j.chb.2020.106582>

This study investigates how social presence in online discussion forums correlates with student prestige in Massive Open Online Courses (MOOCs). The authors develop an automated text classification model using deep learning to analyze forum posts for social presence indicators, such as asking questions, expressing gratitude, and sharing resources. Additionally, they use social network analysis (SNA) to measure learners' prestige via in-degree centrality and authority scores. The findings reveal that learners who engage in positive social behaviors, such as asking questions and sharing insights, tend to receive more responses and establish stronger social connections. Notably, negative emotions and disagreements were linked to lower engagement and prestige in the network. A significant methodological contribution of this study is the implementation of BERT-based text classification, which outperformed traditional machine learning models in analyzing student interactions. The study suggests

that deep learning models, particularly BERT, can be effectively applied to analyzing discussion forums to improve online engagement and learning outcomes.

- Specific linguistic features can predict engagement, which can be used for refining LDA, BERT, and TF-IDF models for dataset

Dyulicheva, Y. Y. (2021). *Learning Analytics in MOOCs as an Instrument for Measuring Math Anxiety. Educational Studies Moscow, 4*, 243–265. <https://doi.org/10.17323/1814-9545-2021-4-243-265>

This study applies BERT and k-means clustering to analyze sentiment in math-related MOOC discussions with the goal of identifying math anxiety patterns among learners. The research collected 1,898 student reviews from Udemy MOOCs and employed VADER sentiment analysis to detect negative sentiments in math-related comments. Using BERT embeddings and k-means clustering, the authors identified distinct emotion-based clusters, including frustration with math, regret over past experiences, and gradual improvement in confidence. The study also constructs a knowledge graph that visualizes relationships between student emotions and their attitudes toward math learning. The authors highlight that BERT's contextualized embeddings outperform traditional text vectorization methods, stating that “representation of reviews as dense vectors allows for a more meaningful clustering of emotional states.” The research suggests that automated sentiment analysis and clustering could be applied to detect learning barriers and improve student engagement in online courses.

[Authors Unknown]. (2020). *Deep Sentiment Classification and Topic Discovery on Novel Coronavirus or COVID-19 Online Discussions NLP Using LSTM Recurrent Neural Network Approach*.

This study examines LSTM-based sentiment classification and topic discovery in large-scale COVID-19 online discussions. The researchers utilize LDA (Latent Dirichlet Allocation) for topic modeling and LSTM recurrent neural networks for sentiment classification. Their findings highlight that LSTM models outperform traditional classifiers in capturing sentiment shifts within lengthy discussions. The study also discusses preprocessing steps such as stopword removal, tokenization, and stemming, making the methodology replicable for other discussion-based datasets. This research is relevant to my project as it offers a deep learning-based sentiment analysis framework that can be adapted for student discussion forums. Given its focus on text-based sentiment detection and topic modeling, this study provides a foundation for comparing LSTM-based classification with transformer models like BERT.

- it provides a replicable sentiment classification framework and topic discovery model for online discussions.

[Authors Unknown]. (2022). *Application of NLP and Deep Learning in Online Education*. *Computers & Education*, 174, 1-14. <https://doi.org/10.1016/j.compedu.2022.103712>

This paper explores the role of NLP and deep learning techniques in online education, particularly for automated discussion analysis, sentiment classification, and engagement tracking. The study compares traditional NLP approaches (TF-IDF, LDA) with deep learning models (BERT and GPT-based transformers), highlighting that fine-tuned BERT classifiers outperform conventional NLP methods in distinguishing nuanced emotional states in online discussions. The authors also discuss feature extraction techniques and preprocessing methods, making the study highly replicable. This research supports my work by validating BERT-based sentiment classification as an effective method for analyzing student engagement in discussion forums and provides insights into feature engineering for deep learning models.

- validates BERT's effectiveness for sentiment classification and provides a comparison between traditional NLP and deep learning models.

Farshad, S., Zorin, E., Amangeldiuly, N., & Fortin, C. (2024). *Engagement Assessment in Project-Based Education: A Machine Learning Approach in Team Chat Analysis*. *Education and Information Technologies*, 29, 13105-13131. <https://doi.org/10.1007/s10639-023-12381-5>

This study presents a machine learning approach to engagement detection in project-based learning, focusing on team chat analysis. The researchers employ Gradient Boosting Models (GBM) to classify engagement levels, using features such as message frequency, sentiment scores, and interaction patterns. The study finds that GBM models provide a computationally efficient alternative to deep learning while maintaining high accuracy in engagement classification. This research is useful for my project as it offers a non-deep-learning alternative for engagement prediction, which can be compared to transformer-based models like BERT. Additionally, it introduces feature engineering techniques that can enhance engagement analysis in online discussion forums.

- alternative engagement classification approach (GBM) that is efficient and easily replicable.

[Authors Unknown]. (2023). *Sentiment Detection for Online Learning Discussions Using NLP*. *Journal of Learning Analytics*, 11(3), 82-105. <https://doi.org/10.18608/jla.2023.8403>

This study investigates sentiment analysis methods for student discussions in online learning platforms, comparing TF-IDF, LDA, and BERT-based classifiers. The results show that BERT outperforms traditional NLP models in detecting sentiment trends over time, while TF-IDF and LDA remain effective for extracting key discussion topics. The study also includes detailed preprocessing techniques, such as lemmatization, tokenization, and stopword removal, making it easily replicable for other educational datasets. This research directly aligns with my project as it provides empirical evidence supporting BERT's effectiveness in sentiment classification, allowing me to compare its performance with TF-IDF and LDA models in my own dataset.

- validates BERT's role in sentiment classification and provides replicable preprocessing steps.

Belford, R., Mac Namee, B., & Greene, D. (2017).

A synthetic dataset generator for topic modelling. In Proceedings of the 25th AIAI Irish Conference on Artificial Intelligence and Cognitive Science (AICS). CEUR Workshop Proceedings.

https://ceur-ws.org/Vol-2086/AICS2017_paper_36.pdf

This paper introduces a semi-synthetic dataset generator designed to simulate known topic distributions and concept drift, allowing for systematic testing of topic modeling algorithms. The authors use controlled word distributions to mimic real-world behavior while maintaining transparency in ground truth. This is particularly relevant for educational NLP projects where labeled data is limited. The study validates the usefulness of synthetic data for testing and comparing topic models like LDA — directly informing our team's work in creating and validating synthetic discussion data.

Belford, R., Mac Namee, B., & Greene, D. (2017). A synthetic dataset generator for topic modelling.

Proceedings of the 25th AIAI Irish Conference on Artificial Intelligence and Cognitive Science (AICS). CEUR Workshop Proceedings. https://ceur-ws.org/Vol-2086/AICS2017_paper_36.pdf

This paper introduces a semi-synthetic dataset generator that simulates known topic distributions and concept drift, making it easier to test how well different topic modeling algorithms perform. It's useful because it gives a replicable method for generating synthetic corpora, which is super relevant to our team since we've been working with fake discussion posts. The fact that the topics are pre-defined also helps evaluate model accuracy in a way we can't always do with real, messy data. This research validates our use of synthetic data to test and compare LDA models before applying them to real CS 1301 forums.

Berenstein, D. (2024). Fine-tune ModernBERT on synthetic data. *Hugging Face Blog*.

<https://huggingface.co/blog/davidberenstein1957/fine-tune-modernbert-on-synthetic-data>

This Hugging Face blog shows how you can fine-tune BERT models using synthetic datasets generated by GPT-based models. The author demonstrates that even without real-world labeled data, BERT can still be trained effectively using well-constructed synthetic examples. This is helpful for our team because we're using mock forum posts to train models, and this article confirms that that's a legit and scalable way to do it. It also connects directly to our future plans to compare BERT's performance with traditional models like TF-IDF and LDA.

Grootendorst, M. (2023). BERTopic: Topic modeling with BERT embeddings. *BERTopic Documentation & Tutorials*. <https://maartengr.github.io/BERTopic/>

This documentation outlines how BERTopic works by combining BERT embeddings with clustering algorithms and class-based TF-IDF to create interpretable topic models. What makes it useful for us is that it supports online learning, which means it can keep learning as new discussion posts come in—great for educational forums. While we didn't implement BERTopic this semester, it's definitely a next step we could explore to improve the interpretability and flexibility of our topic modeling pipeline.

Hugging Face. (2023). Fine-tune ModernBERT for RAG with synthetic data. *Hugging Face Blog*. <https://huggingface.co/blog/sdiazlor/fine-tune-modernbert-for-rag-with-synthetic-data>

This post goes into how synthetic data can be used to fine-tune BERT models for Retrieval-Augmented Generation (RAG), a model architecture that retrieves info and generates responses. While our project isn't building a chatbot, this kind of synthetic Q&A data setup could help us develop future features like auto-generated feedback for instructors or smart summaries of discussion threads. The article supports the idea that synthetic data can do more than just train classifiers—it can help create entire systems that replicate real student-instructor interaction.

Cummins, J. A., & Nambudiri, V. E. (2022). Natural language processing: A window to understanding skincare trends. *International Journal of Medical Informatics*, 167, 104705. <https://doi.org/10.1016/j.ijmedinf.2022.104705>

This study uses Reddit data from r/SkincareAddiction to identify 25 distinct topics using LDA and tracks how interest in those topics changes over time. They also used VADER sentiment analysis to score how positively or negatively users talked about each category. What's especially helpful for my team is that their methodology almost exactly matches ours—from preprocessing with Gensim and NLTK to building a bag-of-words model and running LDA with the same parameters. Even though their focus is dermatology, the NLP pipeline is totally transferable to educational forums. I can use their visualizations and techniques to help justify how we structured our own synthetic and real data experiments, and apply the same kind of sentiment and trend tracking to help instructors understand student engagement better.

